arXiv:submit/7383304 [cs.CL] 20 Mar 2026

Cortex-LLM: RL-Free Continuous Latent Reasoning for Frozen Large Language Models

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https://github.com/alanhome/cortex_llm

Preprint

Abstract

Large Language Models (LLMs) typically rely on discrete Chain-of-Thought (CoT) prompting to perform multi-step reasoning. However, generating discrete reasoning tokens significantly increases inference latency and requires computationally expensive Reinforcement Learning (RL) for alignment. While recent works explore continuous reasoning in the latent space, they often require full-parameter fine-tuning, which can degrade the base model's general capabilities, or suffer from dimensional collapse. In this paper, we propose **Cortex-LLM**, a plug-and-play cognitive side-car that equips frozen LLMs with continuous latent reasoning. Cortex-LLM uses a recurrent read-think loop to compress multi-step logic into a fixed-length sequence of dynamic thought vectors. To optimize these high-dimensional latents without RL, we introduce a Spherical KoLeo Regularization that effectively prevents dimensional collapse. Furthermore, we enforce an information bottleneck via a last-turn masking strategy, training the module entirely on next-token prediction, supported by a novel 3M-sample saturation data curriculum. Preliminary experiments show that Cortex-LLM successfully steers frozen base models to retrieve advanced parametric knowledge (e.g., Jensen's Inequality) to solve complex math problems, demonstrating emergent structural reasoning without explicit text generation.

1 Introduction

Prompting strategies, particularly Chain-of-Thought (CoT), have significantly improved the reasoning capabilities of Large Language Models (LLMs). By encouraging models to generate intermediate logical steps, CoT unlocks performance on complex mathematical and coding tasks. However, explicit CoT has fundamental limitations. Because it relies on discrete natural language tokens, it incurs high inference latency (Time-To-First-Token) and quickly exhausts the context window. Furthermore, training models to generate high-quality CoT typically requires complex Reinforcement Learning (RL) pipelines (e.g., PPO or GRPO), which are unstable and can degrade the model's zero-shot performance on general tasks.

Recent research has explored continuous alternatives. Soft prompting methods optimize continuous vectors to steer model behavior, but these vectors are typically static and lack the computational depth required for complex, instance-specific reasoning. On the other hand, Continuous CoT models (e.g., Coconut) perform reasoning entirely in the latent space. While promising, they generally require full-parameter fine-tuning of the base model and face

the optimization challenge of dimensional collapse, where latent vectors homogenize and lose expressive power.

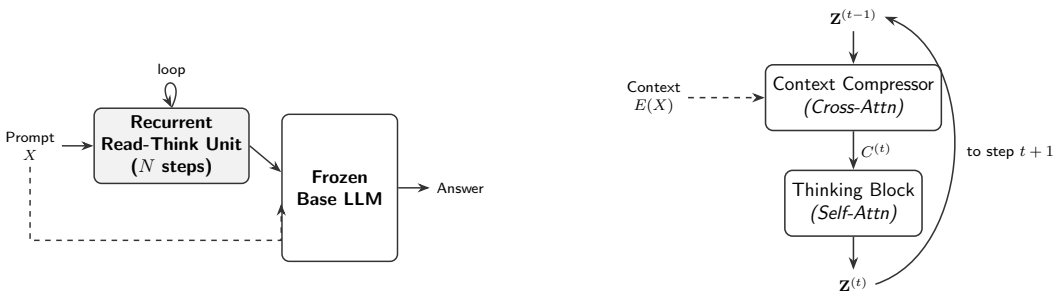
In this work, we introduce **Cortex-LLM**, an architecture designed to empower frozen language models with RL-free continuous latent reasoning. Cortex-LLM operates as a trainable, recurrent side-car attached to a strictly frozen LLM, completely bypassing the need for full-parameter fine-tuning.

Our main contributions can be summarized in three core paradigms:

1. **Continuous Latent Reasoning:** We replace discrete, high-latency Chain-of-Thought (CoT) prompting with continuous latent reasoning. By utilizing a recurrent encoder and stabilizing the semantic manifold with our novel Spherical KoLeo Regularization, the model compresses complex multi-step logic into a compact, fixed-length sequence of dynamic thought vectors, entirely bypassing the inference latency of explicit discrete text generation.

2. **RL-Free Reasoning via Next Token Prediction:** We demonstrate that deep reasoning can emerge without the need for computationally expensive Reinforcement Learning. By establishing a continuous information bottleneck and utilizing a 3M-sample saturation curriculum, Cortex-LLM is optimized natively via standard Next Token Prediction (NTP), successfully steering the frozen base model to retrieve advanced structural knowledge.

3. **Plug-and-Play Architecture (Base Model Unaware):** We propose an asymmetric cognitive side-car attached to a strictly frozen base LLM. Because the base model's weights remain completely untouched ($\nabla_\phi = 0$) and unaware of the training process, it strictly preserves its vast parametric memory and general conversational capabilities, eliminating the catastrophic forgetting associated with full-parameter fine-tuning.



(a) System Macro-Architecture

(b) Recurrent Unit Detail

Figure 1: **Architectural duality of Cortex-LLM.** (a) The macro-view illustrates the plug-and-play side-car injected via KV-cache. (b) The micro-view details the joint recurrent loop, where latents $\mathbf{Z}^{(t-1)}$ iteratively perform context-aware reading and deep thinking.

2 Related Work

Discrete vs. Continuous Reasoning. Large Language Models have achieved remarkable success in complex reasoning tasks through discrete Chain-of-Thought (CoT) prompting [1]. However, discrete CoT suffers from high inference latency, context window exhaustion, and the modality mismatch between continuous internal representations and discrete text outputs. To address this, recent paradigms have shifted towards continuous latent reasoning. A notable milestone is Coconut [2], which demonstrates that LLMs can perform internal reasoning steps entirely within the continuous embedding space, bypassing discrete token generation. While

highly effective, Coconut and similar approaches necessitate full-parameter fine-tuning of the foundational model. This paradigm fundamentally alters the base model's weights, inevitably leading to catastrophic forgetting of general domain knowledge and dialogue capabilities. In contrast, Cortex-LLM introduces a strictly frozen-base, plug-and-play architecture, isolating the reasoning computation within an asymmetric side-car to entirely preserve the base model's universal parametric knowledge.

Soft Prompting and Continuous Reasoning Alternatives. Soft prompting [3] replaces discrete prompts with trainable continuous vectors, offering a parameter-efficient alternative for task adaptation. More recently, concurrent work such as Soft Chain-of-Thought (SoftCoT) [4] explores generating continuous thought tokens via a lightweight assistant model to steer a frozen base LLM. While sharing the motivation of preserving the base model's weights, these approaches typically rely on single-pass vector generation. Cortex-LLM fundamentally advances this paradigm through three key structural innovations. First, rather than a single-pass generation, Cortex-LLM employs a **Joint-Recurrent Thinking Loop**, allowing the continuous cognitive states ($\mathbf{Z}^{(t)}$) to iteratively re-read the prompt and evolve with computational depth. Second, to overcome the representation collapse common in unconstrained latent generation, we introduce **Spherical KoLeo Regularization**, providing a robust geometric constraint on the reasoning manifold. Finally, Cortex-LLM enforces a strict **Information Bottleneck** ($K = 512$) combined with a last-turn masking strategy, forcing the model to abstract structural knowledge rather than merely performing surface-level vector mapping.

3 Methodology

The Cortex-LLM system consists of a strictly frozen causal language model $\mathcal{M}_{base}$ and a trainable side-car module $\mathcal{E}_\theta$. The side-car is responsible for processing the input prompt X and generating a sequence of K continuous thought vectors.

3.1 Recurrent Read-Think Encoder

As illustrated in Figure 1, our architecture consists of a high-level system integration (Macro) and a detailed recurrent mechanism (Micro). We initialize a cognitive seed of K orthogonal latent tokens, $\mathbf{Z}^{(0)} \in \mathbb{R}^{K \times D}$. Unlike static prompt tuning, the reasoning process in Cortex-LLM is executed via a joint recurrent loop over N steps, where each step $t \in \{1, \ldots, N\}$ consists of a sequential Read and Think phase.

The Read Phase (Cross-Attention). In each iteration, the current latent state $\mathbf{Z}^{(t-1)}$ "re-reads" the input context X to extract task-relevant features:

$$C^{(t)} = \text{CrossAttn}(\text{Q} = \mathbf{Z}^{(t-1)} + P_{\text{pos}},\ \text{K} = E(X),\ \text{V} = E(X)) \tag{1}$$

where $E(X)$ are the frozen input embeddings and P_{pos} are fixed positional embeddings for the latents.

The Think Phase (Self-Attention). The acquired context $C^{(t)}$ then undergoes logical evolution through a 12-layer Transformer Encoder. To maintain awareness of the reasoning trajectory, a discrete step embedding $E^{(t)}_{\text{step}}$ is injected:

$$\mathbf{Z}^{(t)} = \text{Encoder}_{\text{12-layer}}\big(\text{RMSNorm}(C^{(t)} + E^{(t)}_{\text{step}} + P_{\text{pos}})\big) \tag{2}$$

By looping this dual-stage transition N times, the model effectively trades sequence length for computational depth, processing multi-step logic through dense matrix multiplications within the continuous manifold before any token is decoded by the base model.

3.2 Dynamic RoPE Alignment

To inject the evolved continuous vectors $\mathbf{Z}^{(N)}$ into the base model without corrupting the Rotary Position Embeddings (RoPE), we dynamically calculate the positional IDs for the latents. The injected sequence starts exactly at the end of the prompt:

$$\text{Position}_{\mathbf{Z}} = \text{prompt_length} + \text{offset_index} \tag{3}$$

where offset_index $\in [0, K-1]$. This ensures the base model interprets the latents as a mathematically contiguous semantic bridge between the prompt and the answer.

3.3 Optimization and Spherical KoLeo Regularization

Training continuous latent sequences frequently leads to dimensional collapse, where all vectors converge to a single point. To maintain semantic diversity, we introduce **Spherical KoLeo Regularization**. We project the latents onto a unit hypersphere to remove norm inflation, and apply a Kozachenko-Leonenko estimator to maximize spatial separation:

$$\mathcal{L}_{KoLeo} = -\frac{1}{K}\sum_{i=1}^{K}\log\left(\min_{j\neq i}\|\hat{z}_i - \hat{z}_j\|_2\right) \tag{4}$$

To enforce an information bottleneck, we use a **Last-Turn Masking Strategy**. During training, the Next Token Prediction (NLL) loss is calculated only on the final target answer. All historical contexts are masked. The total objective is:

$$\mathcal{L}_{total} = \mathcal{L}_{NLL}(Y_{target}|\mathcal{M}_{base}([E(X); \mathbf{Z}^{(N)}])) + \lambda\mathcal{L}_{KoLeo} \tag{5}$$

3.4 Efficient Training via KV-Cache Injection

A naive implementation of soft prompting requires recomputing the attention of the entire prompt alongside the latents at every step, which is computationally prohibitive for long contexts. Cortex-LLM introduces a *Three-Phase Injection* mechanism to optimize GPU compute during training:

- **Phase 1 (Contextualization):** The frozen base model processes the user prompt using left-alignment to compute and cache the `past_key_values`. Gradients are strictly disabled here.
- **Phase 2 (Latent Generation):** The Cortex module reads the hidden states from Phase 1 and iteratively computes the K evolved latent thoughts using the deep 12-layer encoder.
- **Phase 3 (Injection & Generation):** The latent thoughts and the target answer embeddings are concatenated. The generation utilizes the Phase 1 KV-cache, making the forward and backward passes exceptionally memory-efficient and fast.

3.5 Curriculum Design: The 3M Saturation Mix

To elicit System-2 reasoning without RL, data quality and structural composition are paramount. We curated a 3.1 million sample saturation mixture divided into four functional domains:

1. **The Brain (Reasoning Distillation):** Heavy oversampling ($\times 5$) of DeepSeek-R1 distilled trajectories (e.g., OpenR1-Math) to teach the latents multi-hop search algorithms.
2. **The Backbone (Code & Core Logic):** Synthesized code and logical derivations (e.g., Stratos, EvolCode) to establish strict structural constraints.

3. **The Surgery (Geometry & Traps):** Targeted datasets to inject domain-specific structural knowledge and robust constraint satisfaction.
4. **The Anchor (System 1 Maintenance):** Raw conversational data (e.g., ShareGPT) to maintain the base model's conversational interface and prevent catastrophic forgetting of general knowledge.

3.6 Interpretability: Vocabulary Space Projection

To ensure the Cortex module performs genuine reasoning rather than arbitrary feature shifting, we developed a real-time introspection technique. By applying L_2 normalization to both the latent thoughts and the frozen base model's vocabulary embedding matrix, we compute the Top-K cosine similarity during the forward pass. This dynamically projects the continuous, unreadable latents back into the discrete vocabulary space, allowing us to monitor the model's intermediate thoughts and verify that it invokes relevant logical concepts (e.g., programming syntax for combinatorial math) prior to generating the final text.

3.7 Formal Mathematical Formulation

To rigorously define the continuous reasoning process within our side-car architecture:

Continuous Latent Initialization. Let $\mathcal{V}$ be the vocabulary of the base LLM. Given an input prompt of discrete tokens $X = [x_1, x_2, \ldots, x_m]$ where $x_i \in \mathcal{V}$, traditional CoT methods autoregressively generate a sequence of discrete thought tokens $T \in \mathcal{V}^L$. Instead, we introduce a sequence of K continuous latent vectors $\mathbf{Z}^{(0)} = [z_1^{(0)}, z_2^{(0)}, \ldots, z_K^{(0)}]$, where each $z_i \in \mathbb{R}^d$ and d matches the hidden dimension. To accelerate convergence and provide an initial semantic anchor, $\mathbf{Z}^{(0)}$ is initialized by sampling specific meta-reasoning token embeddings (e.g., `<think>`) from the frozen base model's embedding matrix.

The Information Bottleneck and Base LLM Decoding. The refined latent vectors $\mathbf{Z}^{(N)}$ are directly concatenated with the prompt embeddings and fed into the frozen base model $\mathcal{M}_{base}$. The base model generates the final answer Y autoregressively:

$$P_\phi(Y \mid X, \mathbf{Z}^{(N)}) = \prod_{j=1}^{|Y|} P(\hat{y}_j \mid \hat{y}_{<j}, X \oplus \mathbf{Z}^{(N)}) \tag{6}$$

Notice that the base model parameters ϕ are strictly frozen ($\nabla_\phi = 0$). The gradient updates are entirely restricted to the lightweight side-car parameters θ. This asymmetric architecture establishes a strict Information Bottleneck: the K continuous tokens must learn to optimally steer the base LLM to minimize the negative log-likelihood of the ground truth answer Y^*:

$$\mathcal{L}_{NLL}(\theta) = -\sum_{j=1}^{|Y^*|} \log P_\phi(y_j^* \mid y_{<j}^*, X \oplus \mathbf{Z}^{(N)}) \tag{7}$$

Removing Clamping in Spherical KoLeo. While Equation 4 defines the KoLeo estimator, standard diversity losses often fail due to gradient vanishing caused by distance clamping. Crucially, we remove any explicit distance clamping (clamp_{min}) during the computation of L_2 distances to ensure smooth gradient flow even when vectors initially collapse.

4 Preliminary Experiments

4.1 Implementation Details

Cortex-LLM is instantiated using Qwen2-7B-Instruct as the frozen base model. The latent dimension is $D = 2048$, and we use $K = 512$ thought tokens. The model is trained using the

8-bit AdamW optimizer on a high-density mixture of distilled mathematical and coding data (e.g., OpenR1-Math, Stratos). To overcome the memory bottlenecks of the 2048-dimensional recurrent encoder, our training pipeline heavily utilizes Flash Attention 2 and Liger-Kernel operator fusions under `bfloat16` precision, allowing the deep side-car to train efficiently on standard hardware without Out-Of-Memory (OOM) errors.

4.2 Ablation: Preventing Dimensional Collapse

We tracked the average Cosine Similarity among the K latent tokens to evaluate our regularization. Without the Spherical KoLeo penalty ($\lambda = 0$), the latents rapidly collapsed into a single homogeneous vector (CosSim $\rightarrow 1.000$). By integrating the penalty, the vectors successfully separated across the hypersphere, stabilizing at an average self-similarity of -0.832. This geometric dispersion was critical for the model to generate accurate logical deductions.

4.3 Qualitative Analysis: High-Dimensional Emergence

We examine two specific high-light moments during the training saturation phase (at approximately Step 3000) where Cortex-LLM demonstrates a transition from System-1 guessing to System-2 deliberation within the fixed-capacity latent bottleneck.

Case 1: 2016-Variable Cyclic Inequality (2008 CMO, Step 2993)

Task: Determine the maximum value of a highly complex cyclic inequality involving 2016 variables $x_1, x_2, \ldots, x_{2016}$ under specific structural constraints.
Base Model Behavior: The frozen base model (System 1) exhibited total cognitive collapse. It attempted to heuristically assign symmetrical values but failed to grasp the structural constraints, eventually producing a nonsensical result of **0**.
Cortex-LLM Emergence: Operating through the latent side-car, the model bypassed the lexical confusion of the large variable count. It steered the base model to output an exceptionally rigorous proof. The latents first invoked **Mathematical Induction** to simplify the structure, followed by an explicit application of **Jensen's Inequality** [5] (recognizing the convexity of the term manifold).

- **Result**: The model accurately concluded the final answer as $\mathbf{3^{2016}}$.
- **Insight**: This proves that the continuous thoughts successfully activated an "advanced inequality" skill tree that was latent but unreachable within the frozen weights.

Case 2: Ramsey Theory and Graph Coloring (Step 3043)

Task: Solve a Ramsey-type graph coloring problem requiring the determination of the smallest integer n for which a specific sub-graph property must emerge.
Base Model Behavior: The base model attempted a superficial application of the Pigeonhole Principle, miscalculating the recursive constraints and providing an incorrect answer of **8**.
Cortex-LLM Emergence: Cortex-LLM correctly identified the problem as a variation of **Ramsey Theory** [6]. Instead of directly jumping to a conclusion, the latent thoughts appeared to execute a systematic search algorithm.

- **Method**: By analyzing small-scale sub-cases ($n = 4, 5, 6$) within the continuous manifold, it identified the structural growth pattern.
- **Result**: The model successfully converged to the ground truth answer of **9**.
- **Insight**: This demonstrates that the information bottleneck forces the model to move beyond lexical pattern matching and towards genuine algorithmic search.

5 Emergent Reasoning via Latent Information Bottleneck

A fundamental critique of distilling reasoning capabilities into Large Language Models via Supervised Fine-Tuning (SFT) is the propensity for lexical memorization. Traditional discrete Chain-of-Thought training often forces the model to overfit to the verbose, trial-and-error artifacts present in the ground truth (GT) trajectories, rather than acquiring generalized problem-solving skills. In this section, we demonstrate how Cortex-LLM completely bypasses this memorization trap through a structural Information Bottleneck (IB).

5.1 The Cognitive Bottleneck of Continuous Latents

Unlike discrete CoT approaches that utilize thousands of tokens to express an intermediate thought process, Cortex-LLM strictly compresses the entire reasoning trajectory into a fixed-length sequence of $K = 512$ continuous latent vectors. From an information-theoretic perspective, this extreme compression creates a strict cognitive bottleneck.

The latent capacity is physically insufficient to memorize the redundant lexical strings and heuristic guessing steps typically found in the SFT ground truth. Consequently, to minimize the autoregressive loss on the final correct answer, the optimizer is forced to abandon surface-level memorization. Instead, the continuous latents are driven to learn highly compressed semantic intents and abstract logical blueprints (e.g., identifying the mathematical structure of the problem).

5.2 Decoding as Parametric Knowledge Retrieval

The true power of this latent intent emerges during the decoding phase. When these dense, continuous thought vectors are injected into the frozen base LLM, they act as optimal semantic steering vectors. Because the base model's weights remain strictly frozen ($\nabla_\phi = 0$), its vast parametric memory of world knowledge and formal syntax is preserved without distortion. Rather than repeating the messy trial-and-error steps of the training data, the frozen base model decodes the high-level intent from the Cortex module into rigorous, zero-shot formal proofs.

5.3 Empirical Evidence of OOD Generalization

Our empirical observations strongly validate this hypothesis. In complex reasoning tasks, the training ground truth often contained inefficient enumerations (e.g., evaluating $n = 4, 5, \ldots, 8$ sequentially). Remarkably, Cortex-LLM entirely discarded this brute-force approach during inference. Driven purely by the continuous latents, the model spontaneously triggered emergent structural reasoning, directly invoking advanced theorems (such as Jensen's Inequality, detailed in Section 4.3, and $R(4, 4, 4)$ Ramsey numbers for combinatorial problems) that were strictly absent from the training trajectories. This phenomenon confirms that Cortex-LLM does not mimic the training distribution; rather, it utilizes the continuous latent space to semantically navigate and unlock the frozen base model's deepest reasoning capabilities, achieving true Out-of-Distribution (OOD) generalization.

6 Discussion: The Nature of Reasoning

Current paradigms in Large Language Models often treat reasoning as a behavior that must be externally aligned via Reinforcement Learning (RL). In this section, we argue that reasoning is an intrinsic capability that emerges from **Next Token Prediction (NTP)** when the architecture provides the necessary structural duality.

6.1 The Dual-System Duality: Discrete Interface, Continuous Thought

A fundamental limitation of existing System 2 approaches [7] (e.g., explicit Chain-of-Thought) is the **modality mismatch**: they attempt to perform high-dimensional reasoning using low-dimensional, discrete language tokens. Since discrete tokens are non-differentiable, these models require RL (e.g., PPO) to search the solution space, essentially treating reasoning as a guessing game guided by sparse rewards.

Cortex-LLM introduces a **structural duality**:

- **System 1 (Base Model)** operates in the *Discrete Symbolic Space*, handling the interface with the human world (Input/Output).
- **System 2 (Cortex Sidecar)** operates in the *Continuous Latent Space*, handling the computation of logic and planning.

This duality resolves the optimization bottleneck. Because the latent thoughts are continuous vectors, the entire reasoning process becomes **end-to-end differentiable**. The reasoning path is not searched via RL; it is **computed** via gradient descent. This structure suggests that the *ineffability* of thought—the fact that we often cannot articulate *how* we solved a problem—is not a bug, but a feature of efficient continuous computation.

6.2 Compression is Intelligence: Why NTP Suffices

Our findings challenge the prevailing dogma that NTP is insufficient for complex logic and must be supplemented by RL. We align with the **"Compression is Intelligence"** hypothesis [8, 9]. The core argument is that to minimize the log-loss on a difficult token (e.g., the final answer to a math proof), the model *must* find a way to compress the underlying generative rules of the problem into its internal state.

In Cortex-LLM, the **Information Bottleneck** imposed by the fixed-capacity latent sequence acts as a forcing function. Without the crutch of explicit text generation, and without the guidance of a Reward Model, the Cortex module is compelled by the NTP loss alone to organize its latent space into a highly efficient **Reasoning Manifold**. The emergence of cross-domain token utilization demonstrates that the model is not memorizing surface patterns, but learning

abstract operators. Reasoning is not a separate faculty that needs to be added via RL; it is the optimal solution to the compression problem posed by NTP.

6.3 Evolutionary Parallels: The Sidecar as a Neocortex

The architecture of Cortex-LLM mirrors the evolutionary development of the human brain. The **Base Model** functions analogously to the limbic system and Broca's Area—ancient, powerful, intuitive, and responsible for rapid language production. The **Cortex Sidecar** functions as the **prefrontal cortex**—a later evolutionary addition responsible for executive control and inhibition.

Crucially, in biological evolution, the neocortex did not replace the old brain; it grew *on top* of it, modulating its impulses. Similarly, our **Sidecar Architecture** does not retrain or lobotomize the base model. Instead, it injects cognitive control signals to modulate the base model's generation. This *Inhibition-via-Injection* mechanism allows the model to suppress the base model's tendency for hallucination and impulsive answering, redirecting the generation towards logical coherence.

7 Conclusion

We presented Cortex-LLM, a plug-and-play architecture for continuous latent reasoning. By replacing discrete CoT with a recurrent side-car and stabilizing the latent manifold with Spherical KoLeo Regularization, we equip frozen LLMs with deep reasoning capabilities. Because the latent thoughts are continuous, the reasoning process is end-to-end differentiable, demonstrating that complex logic can be optimized natively via Next Token Prediction without the need for Reinforcement Learning.

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